Inverted Single Pendulum

Shygaan Khalili

October 17, 2026

Abstract

This document outlines the derivation of the equations of motion for an inverted pendulum on a cart using Lagrangian mechanics.

1 System Definition and Kinematics

Define:
- M: mass of carriage
- m : mass of pendulum
- l : length of pendulum
- θ angular position of pendulum
- $\dot{\theta}$ angular velocity of pendulum
- $\ddot{\theta}$: angular acceleration of pendulum
- x : carriage position
- $\dot{x}$ carriage velocity
- $\ddot{x}$: carriage acceleration

Position of pendulum

$$x_p = x - l\sin\theta$$
$$y_p = l\cos\theta$$

$$\dot{x_p} = \dot{x} - l\dot{\theta}\cos\theta$$
$$\dot{y_p} = -l\dot{\theta}\sin\theta$$

2 Energy Formulation

Kinetic Energy

$$T = \frac{1}{2}m(v_p^2 + v_2)$$

$$\begin{aligned} v_p^2 &= \dot{x_p}^2 + \dot{y_p}^2 \\ &= (\dot{x} - l\dot{\theta}\cos\theta)^2 + (-l\dot{\theta}\sin^2\theta) \\ &= \dot{x}^2 - 2\dot{x}l\dot{\theta}\cos\theta + l^2\dot{\theta}^2\cos^2\theta + l^2\dot{\theta}^2\sin^2\theta \\ &= \dot{x}^2 - 2\dot{x}l\dot{\theta}\cos\theta + l^2\dot{\theta}^2(\cos^2\theta + \sin^2\theta) \\ &= \dot{x}^2 - 2\dot{x}l\dot{\theta}\cos\theta + l^2\dot{\theta}^2 \end{aligned}$$

$$v_c^2 = \dot{x}^2$$

$$T = \frac{1}{2}m(\dot{x}^2 - 2\dot{x}l\dot{\theta}\cos\theta + l^2\dot{\theta}^2) + \frac{1}{2}M\dot{x}^2$$

Potential Energy

$$V = mgl\cos\theta$$

3 The Lagrangian

$$L = T - V$$

$$L = \frac{1}{2}(M+m)\dot{x}^2\frac{1}{2}(m)(-2\dot{x}l\dot{\theta}\cos\theta + l^2\dot{\theta}^2) - mgl\cos\theta$$

4 Euler-Lagrange Equations

For a general coordinate q:

$$\frac{d}{dt}(\frac{\partial L}{\partial \dot{q}}) - \frac{\partial L}{\partial q} = 0$$

For a general coordinate vector $\vec{q}$:

$$\frac{d}{dt}(\vec{\nabla}_{\dot{q}}L) - \vec{\nabla}_q L = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

Let $\vec{q} = \begin{bmatrix} x \\ \theta \end{bmatrix}$.

$$\vec{\nabla}_{\dot{q}}L = \begin{bmatrix} \frac{\partial L}{\partial \dot{x}} \\ \frac{\partial L}{\partial \dot{\theta}} \end{bmatrix}$$

$$\frac{\partial L}{\partial \dot{x}} = (M+m)\dot{x} - ml\dot{\theta}\cos\theta$$

$$\frac{\partial L}{\partial \dot{\theta}} = -m\dot{x}l\cos\theta - ml^2\dot{\theta}$$

$$\vec{\nabla}_{\dot{q}}L = \begin{bmatrix} (M+m)\dot{x} - ml\dot{\theta}\cos\theta \\ -m\dot{x}l\cos\theta - ml^2\dot{\theta} \end{bmatrix}$$

$$\vec{\nabla}_q L = \begin{bmatrix} \frac{\partial L}{\partial x} \\ \frac{\partial L}{\partial \theta} \end{bmatrix}$$

$$\frac{\partial L}{\partial x} = 0$$

$$\frac{\partial L}{\partial \theta} = m\dot{x}l\dot{\theta}\sin\theta + mgl\sin\theta$$

$$\vec{\nabla}_q L = \begin{bmatrix} 0 \\ m\dot{x}l\dot{\theta}\sin\theta + mgl\sin\theta \end{bmatrix}$$

$$\frac{d}{dt}(\vec{\nabla}_{\dot{q}}L) = \begin{bmatrix} (M+m)\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta \\ -m\ddot{x}l\cos\theta + m\dot{x}l\dot{\theta}\sin\theta - ml^2\ddot{\theta} \end{bmatrix}$$

$$\frac{d}{dt}(\vec{\nabla}_{\dot{q}}L) - \vec{\nabla}_{q}L = \begin{bmatrix} (M+m)\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta \\ -m\ddot{x}l\cos\theta - ml^2\ddot{\theta} - mgl\sin\theta \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} M+m & -ml\cos\theta \\ -ml\cos\theta & -ml^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} ml\dot{\theta}^2\sin\theta \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -ml^2\ddot{\theta} \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

5 Matrix Form and Linearization

$$\begin{bmatrix} M+m & -ml\cos\theta \\ -ml\cos\theta & -ml^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} ml\dot{\theta}^2\sin\theta \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -ml^2\ddot{\theta} \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$